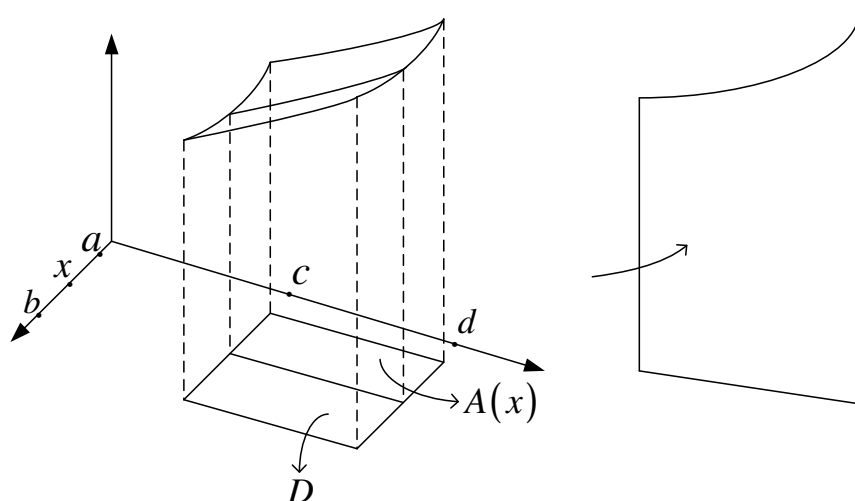


§15.2 Iterated Integrals

Homework : 1,5,11,15,17,21,23,25,29,36

Double Integral : 二維計算屋子體積的方法

Iterated Integrals : 一維計算屋子體積的方法(要算 2 次).



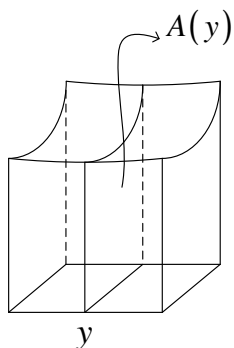
$$(i) \int_c^d f(x, y) dy = A(x)$$

= 固定 x ，屋子的縱裁面面積

$$\int_a^b A(x) dx = \text{將所有縱裁面面積連續相加}$$

$$= \int_a^b \left(\int_c^d f(x, y) dy \right) dx = \int_a^b \int_c^d f(x, y) dy dx$$

= 一維算法算 2 次



同理可得，先固定 y 的縱裁面面積算法而

$$\text{得 } \int_c^d \int_a^b f(x, y) dx dy$$

Problem : 什麼樣的屋頂 ($f(x, y) = z$) 此三種屋子體積算法相等。

Theorem : (Fubini :) If f is continuous on D (屋頂沒破), then

$$\iint_D f(x, y) dA = \int_a^b \int_c^d f(x, y) dy dx = \int_c^d \int_a^b f(x, y) dx dy.$$

Remark :

- (i) In fact, f 只有些許小裂縫, 三種體積算法也是一樣。
- (ii) Clairaut's Theorem : When the order of partial differentiation does not matter?

Fubini Theorem : When the order of iterated integrals does not matter?

Example 1 : Find $\iint_R y \sin(xy) dA : R = [1, 2] \times [0, \pi]$

$$\begin{aligned} \text{Solution : } \iint_R y \sin(xy) dA &= \int_0^\pi \int_1^2 y \sin(xy) dx dy \\ &= \int_0^\pi \left[-\cos xy \right]_1^2 dy \\ &= \int_0^\pi -\cos 2y + \cos y dy \\ &= -\frac{1}{2} \sin 2y \Big|_0^\pi + \sin y \Big|_0^\pi \\ &= 0 \text{ (有地下室, 體積相消.)} \end{aligned}$$

Example 2 : $\iint_R (x - 3y^2) dA \quad R = [0, 2] \times [1, 2]$

Solution :

$$\bullet \int_1^2 \int_0^2 (x - 3y^2) dx dy = \int_1^2 \left(\frac{1}{2} x^2 - 3y^2 x \right) \Big|_0^2 dy = \int_1^2 (2 - 6y^2) dy = -124$$

$$\bullet \int_0^2 \int_1^2 (x - 3y^2) dy dx = \int_0^2 \left(xy - 9y^3 \right) \Big|_1^2 dx = -124$$

Example 3 :
$$\begin{cases} x^2 + 2y^2 + z = 16 \\ x = 2 \\ y = 2 \\ \text{Three coordinate planes} \end{cases} \quad \text{所圍的物體體積.}$$

Solution :

$$\text{屋頂 : } z = 16 - x^2 - y^2$$

$$\text{地基 : } [0, 2] \times [0, 2]$$

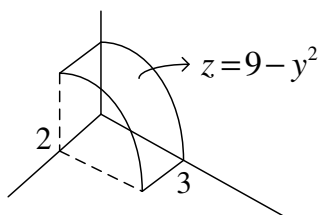
$$V = \int_0^2 \int_0^2 (16 - x^2 - y^2) dx dy = 48$$

Example 4 : Find the volume of the solid in the first octant bounded by the cylinder $z = 9 - y^2$ and the plane $x = 2$.

Solution :

屋頂： $z = 9 - y^2$

地基： $[0, 2] \times [0, 3]$



$$V = \int_0^3 \int_0^2 (9 - y^2) dx dy = 36.$$

Example 5 : If f is continuous on $[a, b] \times [c, d]$ and $g(x, y) = \int_a^x \int_c^y f(s, t) dt ds$

for $a < x < b$, $c < y < d$. Prove that $g_{xy} = g_{yx} = f(x, y)$.

Proof :

$$g_x \stackrel{FTC}{=} \int_c^y f(x, t) dt \Rightarrow g_{xy} \stackrel{FTC}{=} f(x, y).$$

$$g(x, y) \stackrel{Fubini}{=} \int_c^y \left(\int_a^x f(s, t) ds \right) dt$$

$$\Rightarrow g_y \stackrel{FTC}{=} \int_a^x f(s, y) ds \Rightarrow g_{yx} \stackrel{FTC}{=} f(x, y).$$